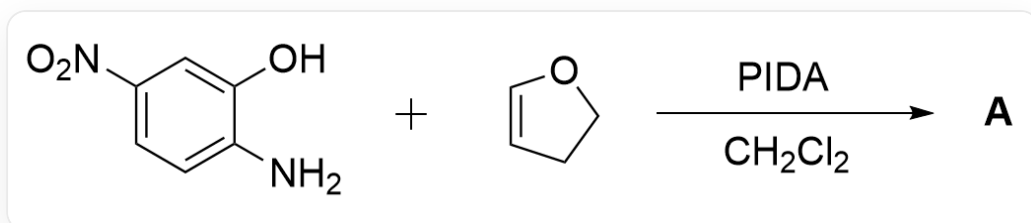


Question

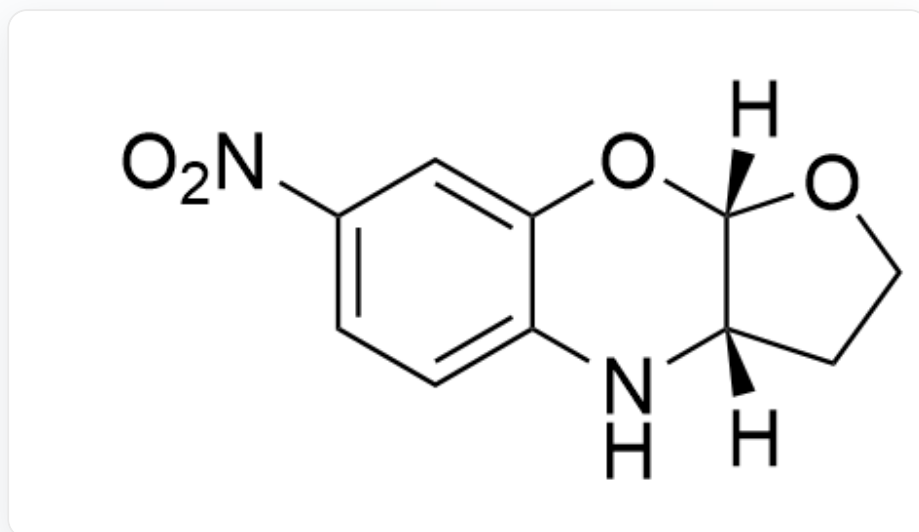


OC1=CC([N+])([O-])=O=CC=C1N.C2=CCCO2>[PIDA].ClCCl>[A], A is the product

Given that the molecular formula of product **A** is $\text{C}_{10}\text{H}_{10}\text{N}_2\text{O}_4$, and it contains 3 rings. Without considering enantiomers, provide the structural formula of **A**.

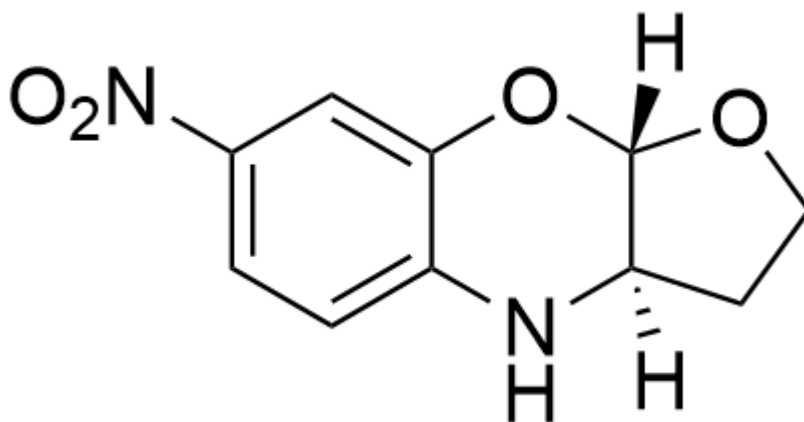
A. All other options are incorrect

B.



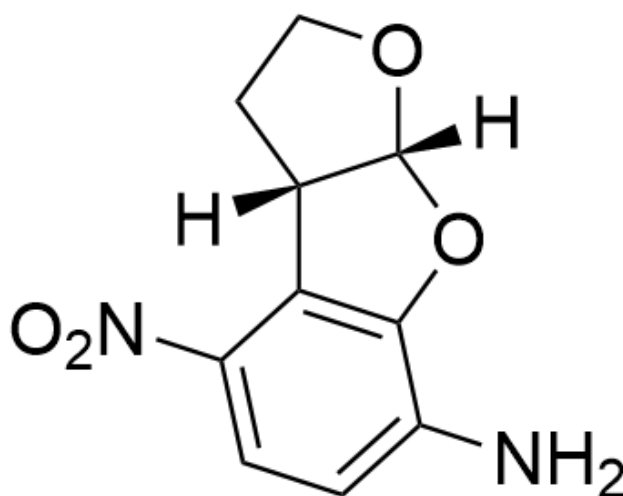
[H][C@]12OC3=CC([N+])([O-])=O=CC=C3N[C@@]1([H])CCO2

C.



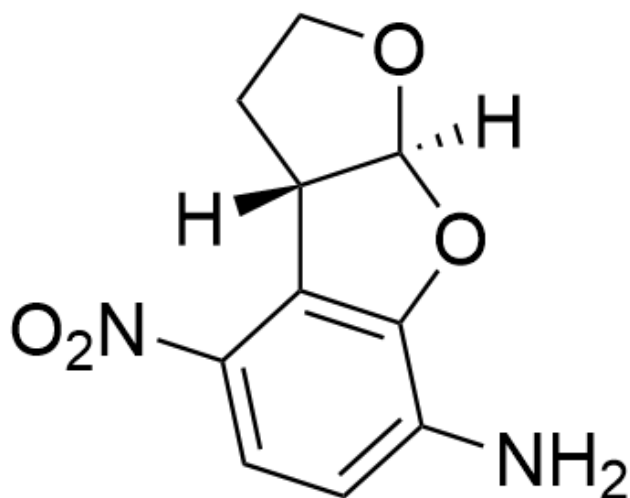
[H][C@]12OC3=CC([N+](O-)=O)=CC=C3N[C@]1([H])CCO2

D.



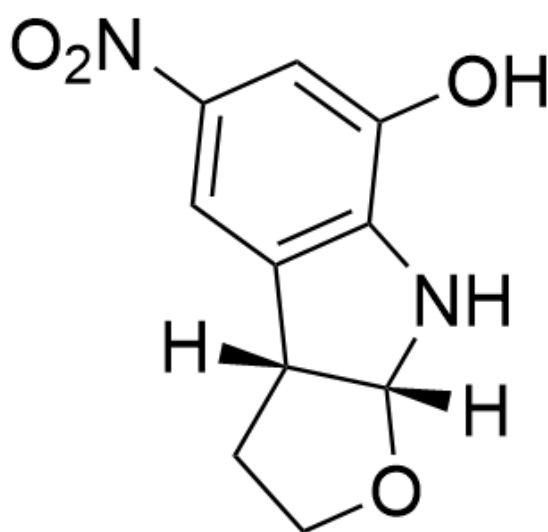
NC1=CC=C([N+](O-)=O)C2=C1O[C@]3([H])[C@@]2([H])CCO3

E.



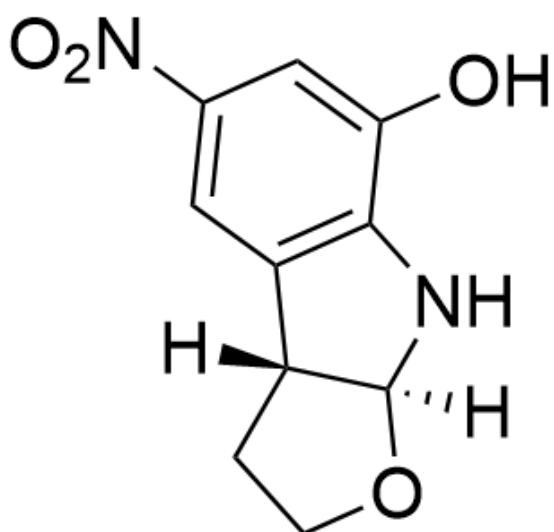
NC1=CC=C([N+](O-)=O)C2=C1O[C@@]3([H])[C@@]2([H])CCO3

F.



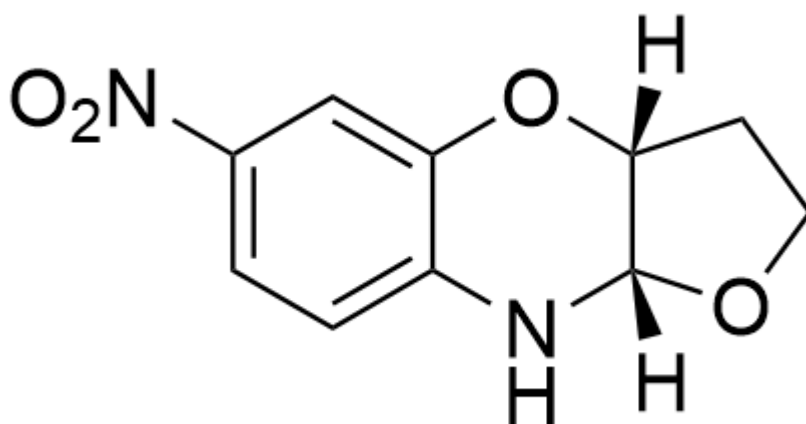
OC1=CC([N+](O-)=O)=CC2=C1N[C@@]3([H])[C@]2([H])CCO3

G.



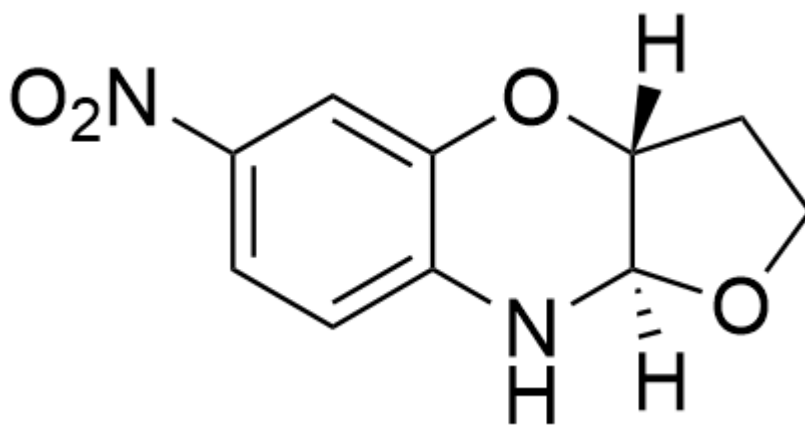
OC1=CC([N+])([O-])=CC2=C1N[C@]3([H])[C@]2([H])CCO3

H.



[H][C@]12OC3=CC([N+])([O-])=CC=C3N[C@@]1([H])OCC2

I.



[H][C@]12OC3=CC([N+](O-)=O)=CC=C3N[C@]1([H])OCC2

Answer

Correct Answer: **B**

Detailed Explanation

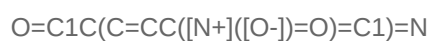
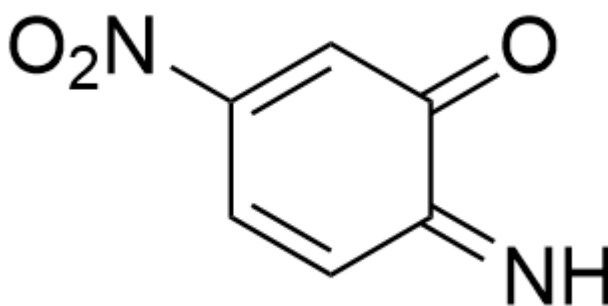
First, based on the molecular formula $C_{10}H_{10}N_2O_4$ of product **A**, it can be deduced that the reaction consumes 1 equivalent of PIDA

CHECKPOINT

1 PTS

First, based on the molecular formula $C_{10}H_{10}N_2O_4$ of product **A**, it can be deduced that the reaction consumes 1 equivalent of PIDA

First, PIDA oxidizes the substrate to obtain an intermediate.



CHECKPOINT

1 PTS

PIDA Oxidation intermediate: O=C1C(C=CC([N+])([O-])=O)=C1=N

Then, the intermediate undergoes an inverse electron-demand D-A reaction with an electron-rich diene to obtain the product.

CHECKPOINT

1 PTS

Then, the intermediate undergoes an inverse electron-demand D-A reaction with an electron-rich diene to obtain the *cis*-product.

For the diene, the electronegativity of O is greater than that of N, so the N-terminus is relatively more electron-deficient.

CHECKPOINT

1 PTS

For the diene, the electronegativity of O is greater than that of N, so the N-terminus is relatively more electron-deficient.

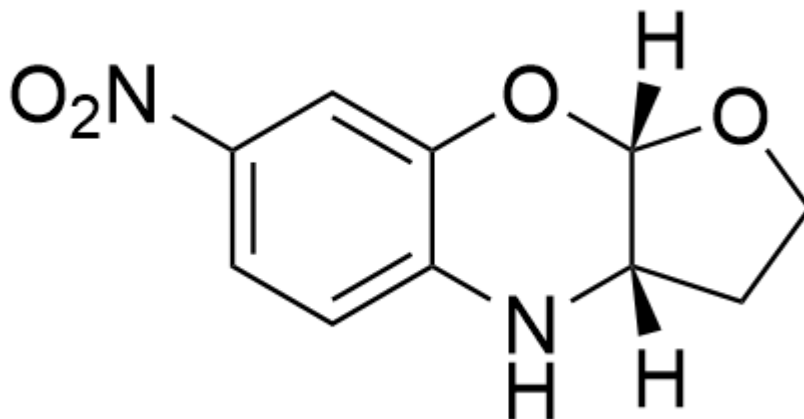
For the dienophile, obviously the C at the *meta*-position of O is more electron-rich.

CHECKPOINT

1 PTS

For the dienophile, obviously the C at the *meta*-position of O is more electron-rich.

Therefore, according to the electronegativity, options H and I can be excluded.



[H][C@]12OC3=CC([N+](O-)=O)=CC=C3N[C@@]1([H])CCO2

CHECKPOINT

1 PTS

Product **A** : [H][C@]12OC3=CC([N+](O-)=O)=CC=C3N[C@@]1([H])CCO2